



Prayas 2.0 2027 Lecture Planner - Physical Chemistry (GOAT)

S. No.	Batch Name	Subject	Sub- Subject	Chapter Name	Topic	Lecture Number	Date	Faculty Name
1	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Nature of Matter, Classification of Matter, Properties of Matter and their Measurement, International System of Units (S.I)	1	Monday, June 8, 2026	Rahul Dudi Sir
2	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Uncertainty in Measurement	2	Tuesday, June 9, 2026	Rahul Dudi Sir
3	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Laws of Chemical Combinations	3	Wednesday, June 10, 2026	Rahul Dudi Sir
4	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Dalton's Atomic Theory, Atomic Mass, Atomic Mass Unit,	4	Thursday, June 11, 2026	Rahul Dudi Sir
5	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Gram Atomic Mass, Average Atomic Mass	5	Friday, June 12, 2026	Rahul Dudi Sir
6	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Molecular Mass, Gram Molecular Mass	6	Saturday, June 13, 2026	Rahul Dudi Sir
7	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Average Molecular Mass Formula Mass	7	Monday, June 15, 2026	Rahul Dudi Sir
8	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Percentage Composition, Empirical and Molecular formula	8	Tuesday, June 16, 2026	Rahul Dudi Sir
9	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Mole, Vapour density, Atomicity Mole Concept	9	Wednesday, June 17, 2026	Rahul Dudi Sir
10	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Concentration Terms & Application	10	Thursday, June 18, 2026	Rahul Dudi Sir
11	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Stoichiometry and Stoichiometric Calculations Gravimetric Analysis	11	Friday, June 19, 2026	Rahul Dudi Sir
12	Prayas 2.0 2027	Chemistry	Physical Chemistry	Some Basic Concepts of Chemistry	Sequential Reaction, POAC, Oleum Labelling Eudiometry	12	Saturday, June 20, 2026	Rahul Dudi Sir
13	Prayas 2.0 2027	Chemistry	Physical Chemistry	Redox Reaction	Introduction	1	Monday, June 22, 2026	Rahul Dudi Sir
14	Prayas 2.0 2027	Chemistry	Physical Chemistry	Redox Reaction	Balancing of Redox Reaction Types of Redox Reaction	2	Tuesday, June 23, 2026	Rahul Dudi Sir
15	Prayas 2.0 2027	Chemistry	Physical Chemistry	Redox Reaction	Concept of n-Factor or Valency Factor	3	Wednesday, June 24, 2026	Rahul Dudi Sir
16	Prayas 2.0 2027	Chemistry	Physical Chemistry	Redox Reaction	Law of Equivalence	4	Thursday, June 25, 2026	Rahul Dudi Sir
17	Prayas 2.0 2027	Chemistry	Physical Chemistry	Redox Reaction	Redox Titration	5	Friday, June 26, 2026	Rahul Dudi Sir
18	Prayas 2.0 2027	Chemistry	Physical Chemistry	Structure of Atom	Iodometric and Iodimetric Titration, Redox Reactions in Electrode process	1	Saturday, June 27, 2026	Rahul Dudi Sir
19	Prayas 2.0 2027	Chemistry	Physical Chemistry	Structure of Atom	Introduction	2	Monday, June 29, 2026	Rahul Dudi Sir
20	Prayas 2.0 2027	Chemistry	Physical Chemistry	Structure of Atom	Fundamental Particles, Discovery of Fundamental Particles Atomic Models	3	Tuesday, June 30, 2026	Rahul Dudi Sir
21	Prayas 2.0 2027	Chemistry	Physical Chemistry	Structure of Atom	Some Important Atomic Terms, Nature of Light and Electromagnetic Spectrum, Particle Nature of Electromagnetic Radiation: Planck's Quantum, Application of Planck's Quantum Theory	4	Wednesday, July 1, 2026	Rahul Dudi Sir
22	Prayas 2.0 2027	Chemistry	Physical Chemistry	Structure of Atom	Bohr's Model for Hydrogen Atom	5	Thursday, July 2, 2026	Rahul Dudi Sir
23	Prayas 2.0 2027	Chemistry	Physical Chemistry	Structure of Atom	Ionization Energy of Hydrogen and Hydrogen-Like Particles	6	Friday, July 3, 2026	Rahul Dudi Sir
24	Prayas 2.0 2027	Chemistry	Physical Chemistry	Structure of Atom	Spectrum: Evidence for the Quantized Electronic Energy Levels	7	Saturday, July 4, 2026	Rahul Dudi Sir
25	Prayas 2.0 2027	Chemistry	Physical Chemistry	Solutions	Spectrum: Evidence for the Quantized Electronic Energy Levels Quantum Mechanical Model of the Atom Quantum Number Rules for Filling of electrons in Orbitals in Atom, Exchange and Pairing energy	1	Wednesday, August 5, 2026	Rahul Dudi Sir
26	Prayas 2.0 2027	Chemistry	Physical Chemistry	Solutions	Binary Solution Concentration Terms	2	Thursday, August 6, 2026	Rahul Dudi Sir
27	Prayas 2.0 2027	Chemistry	Physical Chemistry	Solutions	Solubility Vapour Pressure Vapour Pressure of Liquid Solutions	3	Friday, August 7, 2026	Rahul Dudi Sir
28	Prayas 2.0 2027	Chemistry	Physical Chemistry	Solutions	Vapour Pressure Vapour Pressure of Liquid Solutions Determination of Vapour Pressure of a Liquid Vapour Pressure of Liquid Solutions	4	Saturday, August 8, 2026	Rahul Dudi Sir
29	Prayas 2.0 2027	Chemistry	Physical Chemistry	Solutions	Ideal and Non-ideal Solutions Colligative Properties	5	Monday, August 10, 2026	Rahul Dudi Sir
30	Prayas 2.0 2027	Chemistry	Physical Chemistry	Solutions	Osmosis Osmotic Pressure Abnormal Molar Masses	6	Tuesday, August 11, 2026	Rahul Dudi Sir
31	Prayas 2.0 2027	Chemistry	Physical Chemistry	Solutions	Van't Hoff Factor and Abnormality in Molar Mass	7	Wednesday, August 12, 2026	Rahul Dudi Sir
32	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Kinetics	Abnormal Molar Masses Van't Hoff Factor and Abnormality in Molar Mass	1	Thursday, August 13, 2026	Rahul Dudi Sir
33	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Kinetics	Introduction Rate of Reaction Rate Law Expression Rate constant Factors Affecting the Reaction Rate Order and molecularity of Reaction Negative Order of the Reaction Approximation	2	Friday, August 14, 2026	Rahul Dudi Sir
34	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Kinetics	Integrated Rate Equations Numericals on Some Important First Order Reactions	3	Monday, August 17, 2026	Rahul Dudi Sir
35	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Kinetics	Integrated Rate Equations Pseudo First Order Reaction Experimental Determination of Order of Reaction	4	Tuesday, August 18, 2026	Rahul Dudi Sir
36	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Kinetics	Experimental Determination of Order of Reaction Collision Theory of Chemical Reactions	5	Wednesday, August 19, 2026	Rahul Dudi Sir
37	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Kinetics	Concept of Activation Energy and Transition State Theory Activation Energy Activated Complex Theory	6	Thursday, August 20, 2026	Rahul Dudi Sir

38	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Kinetics	Effect of Catalyst on the Reaction rate Enzyme Catalysis Nuclear Chemistry Radioactive Series and Nuclear Reactions Applications of Radioactivity	7	Friday, August 21, 2026	Rahul Dudi Sir
39	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Introduction,Basic terms,Classification of system on the basis of nature,State of system	1	Saturday, August 22, 2026	Rahul Dudi Sir
40	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Types of properties,TYPES OF THERMODYNAMIC PROCESSES ON BASIS OF THE WAY THE PROCESSES ARE CARRIED OUT Work, heat and internal energy,1st Law of Thermodynamics Enthalpy Heat Capacity	2	Monday, August 24, 2026	Rahul Dudi Sir
41	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	CALCULATION OF Q, W, ΔU & ΔH IN VARIOUS PROCESS	3	Tuesday, August 25, 2026	Rahul Dudi Sir
42	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Thermochemistry Enthalpy of a substance Standard state of substance Enthalpy of reaction Factor affecting heat of reaction LAWS OF THERMO CHEMISTRY	4	Wednesday, August 26, 2026	Rahul Dudi Sir
43	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Enthalpy change for different type of reaction ENTHALPY OF IONIZATION OF WEAK ELECTROLYTE" ENTHALPY OF TRANSITION ENTHALPY OF FORMATION OF IONS Kirchoff's Law	5	Thursday, August 27, 2026	Rahul Dudi Sir
44	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Calorimetry Limitation of 1st law of thermodynamics Spontaneous process Entropy Second law of Thermodynamics ENTROPY CHANGE OF AN IDEAL GAS	6	Saturday, August 29, 2026	Rahul Dudi Sir
45	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Entropy Change of Solids and liquids Upon Heating Entropy Calculation for solid or liquid systems	7	Monday, August 31, 2026	Rahul Dudi Sir
46	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Standard absolute molar entropy	8	Tuesday, September 1, 2026	Rahul Dudi Sir
47	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Comparing absolute entropies of substance	9	Wednesday, September 2, 2026	Rahul Dudi Sir
48	Prayas 2.0 2027	Chemistry	Physical Chemistry	Thermodynamics	Entropy Change (ΔS) of a Chemical Reaction GIBB'S FREE ENERGY	10	Thursday, September 3, 2026	Rahul Dudi Sir
49	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Equilibrium	Introduction,Characteristics of chemical equilibrium Law of mass action	1	Friday, September 4, 2026	Rahul Dudi Sir
50	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Equilibrium	Variation of equilibrium constant with the change of stoichiometric coefficients,Multiple	2	Saturday, September 5, 2026	Rahul Dudi Sir
51	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Equilibrium	Equilibrium Reaction Quotient Equilibrium Constant of Various equilibrium,	3	Monday, September 7, 2026	Rahul Dudi Sir
52	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Equilibrium	Calculation of degree of dissociation from vapour density measurement	4	Tuesday, September 8, 2026	Rahul Dudi Sir
53	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Equilibrium	Le chatelier's principle Effect of temperature on equilibrium constant	5	Wednesday, September 9, 2026	Rahul Dudi Sir
54	Prayas 2.0 2027	Chemistry	Physical Chemistry	Chemical Equilibrium	Free energy and chemical equilibrium Physical Equilibrium	6	Thursday, September 10, 2026	Rahul Dudi Sir
55	Prayas 2.0 2027	Chemistry	Physical Chemistry	Ionic Equilibrium	Type of electrolytes,Different theory for acid and base,The autoionization of water	1	Friday, September 11, 2026	Rahul Dudi Sir
56	Prayas 2.0 2027	Chemistry	Physical Chemistry	Ionic Equilibrium	The pH-Scale Ionisation of weak electrolyte	2	Saturday, September 12, 2026	Rahul Dudi Sir
57	Prayas 2.0 2027	Chemistry	Physical Chemistry	Ionic Equilibrium	Common ion effect and levelling effect	3	Tuesday, September 15, 2026	Rahul Dudi Sir
58	Prayas 2.0 2027	Chemistry	Physical Chemistry	Ionic Equilibrium	pH-Calculation Buffer Solution	4	Wednesday, September 16, 2026	Rahul Dudi Sir
59	Prayas 2.0 2027	Chemistry	Physical Chemistry	Ionic Equilibrium	Salt hydrolysis Indicator	5	Thursday, September 17, 2026	Rahul Dudi Sir
60	Prayas 2.0 2027	Chemistry	Physical Chemistry	Ionic Equilibrium	Acid-base titration Solubility and Solubility product	6	Friday, September 18, 2026	Rahul Dudi Sir
61	Prayas 2.0 2027	Chemistry	Physical Chemistry	Electrochemistry	Introduction Electrochemical Cell Galvanic Cells Liquid Junction Potential or Diffusion Potential Electrochemical Cell electrodes and Electrode potential Electrochemical Series	1	Saturday, September 19, 2026	Rahul Dudi Sir
62	Prayas 2.0 2027	Chemistry	Physical Chemistry	Electrochemistry	Nernst Equation Equilibrium Constant and emf Gibbs Energy of the reaction Application of Nernst equation Various types of half cells Thermodynamics of Galvanic cell Concentration Cell	2	Monday, September 21, 2026	Rahul Dudi Sir
63	Prayas 2.0 2027	Chemistry	Physical Chemistry	Electrochemistry	Electrolytes Electrolytic cell Quantitative Aspects of Electrolysis and Faraday's Laws Metallic/Electrolytic Conductors Determination of Cell Constant	3	Tuesday, September 22, 2026	Rahul Dudi Sir
64	Prayas 2.0 2027	Chemistry	Physical Chemistry	Electrochemistry	Measurement of Conductance and Conductivity of Ionic Solutions Molar Conductivity Equivalent Conductivity	4	Wednesday, September 23, 2026	Rahul Dudi Sir
65	Prayas 2.0 2027	Chemistry	Physical Chemistry	Electrochemistry	Relation Between Molar Conductivity and Equivalent Conductivity Variation of Conductivity and Molar Conductivity with Concentration Kohlrausch's Law	5	Thursday, September 24, 2026	Rahul Dudi Sir
66	Prayas 2.0 2027	Chemistry	Physical Chemistry	Electrochemistry	Absolute Ionic Mobilities and their Calculation Conductometric Titrations Commercial Cells or Batteries	6	Friday, September 25, 2026	Rahul Dudi Sir
67	Prayas 2.0 2027	Chemistry	Physical Chemistry	Electrochemistry	Fuel Cells Hydrogen Economy Corrosion of Metals	7	Saturday, September 26, 2026	Rahul Dudi Sir

68	Prayas 2.0 2027	Chemistry	Physical Chemistry	State of matter	Introduction,Intermolecular Forces The Gaseous State The Gas Laws The Gas Laws,Ideal Gas Equation Dalton's Law of Partial Pressures Graham's Law of Diffusion/effusion Kinetic Molecular Theory of Gases Kinetic Gas Equation Explanation of Gas Law from Kinetic Theory Kinetic Energy and Molecular Speeds Maxwell-Boltzmann Distribution of Molecular Speeds,Collision Theory Collision frequency Mean Free Path Behaviour of Real Gases : Deviation from Ideal Gas Behaviour Equation of State for Real Gases (Van der Waal's Equation) Liquefaction of Gases Joule - Thomson Effect Inversion Temperature Liquid State Vapour Pressure Boiling and Boiling point Viscosity Surface Tension	1	Tuesday, December 29, 2026	Rahul Dudi Sir
69	Prayas 2.0 2027	Chemistry	Physical Chemistry	The Solid State	Introduction Difference between Crystalline and amorphous solids Classification of crystalline solids Types of symmetry in crystals Some basic terms in crystalline solids Cubic unit cell(Bravais lattice) Mathematical analysis of cubic system Close packing of identical solid spheres Voids or interstitial voids Tetrahedral and Octahedral voids in close packed structure (FCC main) Radius ratio rule (For ionic solids) Structure of Ionic Compounds Effect of temperature and pressure on the crystal structure Defects or imperfections in Solids Properties of Solids:Electrical properties Super conducting materials	1	Wednesday, December 30, 2026	Rahul Dudi Sir
70	Prayas 2.0 2027	Chemistry	Physical Chemistry	SURFACE CHEMISTRY	Introduction Adsorption Adsorption Isotherm Application of adsorption Catalysis Colloidal State Type of Colloids Preparation of Colloidal Solution Gels Emulsion Application of Colloids	1	Thursday, December 31, 2026	Rahul Dudi Sir